

Analysis of Herpetofauna Diversity and Trends in Upland Northern Mississippi Hardwood
Forest and Retired Farmland
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Abstract

Species diversity has become of more importance to ecological community surveys and conservation efforts globally. Use of such methods of assessing species in target groups of particular concern, such as herpetofauna, have increased in frequency in recent decades, incorporating mathematical indices and measures in an attempt to analyze community composition as a trend over time. The goal of this study was to report herpetofauna species found in the region, their diversity, dominance and rarity, richness, and other measures throughout the four separate survey trips, showing trends over the periods of time in order to better inform the timing and methods of the survey in future years. Ex-situ data analysis of the detections over four years thus far have found 86 herpetofauna individuals across 26 distinct species. Reptilian species make up the larger portion of detections than amphibian, with a ratio of detections and relative abundance of 1.39:1, respectively. Dominant species across trip years included the southern two-lined salamander, the southern cricket frog, and the little brown skink. Dominant species across seasons included the Mississippi slimy salamander, southern cricket frog, and little brown skink. Highest herpetofauna abundance was found in 2018 and in the fall season, while highest species richness was detected in 2018 and in the spring season. Species evenness was maximized in 2019 and in the fall season. Shannon-Weiner and Simpson's Diversity were both maximized in the 2018 and spring trips and lowest in the 2021 and fall trips overall. Diversity of amphibians in particular was maximized in 2022 and in fall, while diversity of reptiles in particular was maximized in 2018 and in spring season. Reptiles were detected at higher average substrate temperatures and similar air temperatures to amphibians. The highest and lowest temperatures at detection for amphibians were lower than those for reptiles when adjusting for the summer season anomaly found. Deviations from expectations formed from knowledge on herpetological distribution, ecology, and life history were found in herpetofauna abundance and diversity in particular seasons, while other measures of richness, abundance, and evenness met predictions. This survey thus far gave insight into the state of the herpetofauna community in the study region, creating a baseline for comparison in continuation of the study. The value of this extended study thus far will prove critical in moving forward in herpetofauna conservation, particularly with attention to bias and improvements to methods found in the past four years.

Introduction

Biodiversity has become one of the most frequently used evaluation measures for ecological communities, habitats, and their species (Dodd 1992, Iknayan et al. 2014, Martensson 2016). Biodiversity has been defined to include “diversity within species, between species, and of ecosystems” (Martensson 2016). Diversity of species within a particular habitat or compared between habitats or ecosystems has only increased in use within the biological research areas,

arising as a prominent topic in the fields of life-history evolution and community ecology, and has even become a measure indicating population, ecosystem, or environmental health (Dodd 1992, Martensson 2016, Winter et al. 2013). Diversity of species has arisen as a significant aspect of the fundamental definition of community, ecosystem, and ecology, and has been more realized as an important facet in environmental science as a greater study for a multitude of reasons (Martensson 2016). Estimates of species diversity as a parameter, and, hand-in hand, other measures of species richness, abundance, evenness, as trends over time, have become altogether central to community and larger ecosystem ecology, elucidating dynamics that govern ecological communities, linking ecosystem services and keystone, indicator, and signal species, ecosystem function, ecological monitoring, comparative community studies, and spatial community patterns, among others (Iknayan et al. 2014, Laurencio and Fitzgerald 2010, Moritz et al. 2005, Winter et al. 2013). While numerous classes are the focus of diversity-related surveys, herpetofauna are one group highly targeted by such studies and have a growing body of literature surrounding them as of late.

Herpetofauna, or amphibians and reptiles, the focus of the field of herpetology, have been a vertebrate group of particular concern and interest to ecologists and conservation biologists in recent decades (Gibbons and Stangel 1999, Leyte-Manrique et al. 2019, Olson and Saenz 2013, Urbina-Cardona 2008, USGS 2022). These two groups are critical to the maintenance of biological diversity, fulfill numerous functional roles, and are heralds of the environmental quality of the habitats in which they live (Gibbons and Stangel 1999). As of late, these species have garnered concerns surrounding their declining ecological state, along with their habitats. Species of both herpetological classes occupy similar habitats and are equally vulnerable to habitat degradation, their currently known global, anthropogenic threats, to observed and

expected population declines, and to the possibility of extinction (Ruthven et al. 2002, USGS 2022, Whitfield Gibbons et al. 2000). The world's reptiles and amphibians face stressors unique to them with known – climate change, habitat degradation and fragmentation, agricultural intensification, urbanization, pollution, and disease – origins that will only continue to bring about declines, extirpations, and extinctions if not rectified (Ghosh and Basu 2020, Rabbe et al. 2022, USGS 2022, Whitfield Gibbons et al. 2000). Herpetofauna, therefore, have been the focus of much species and community surveying along with conservation efforts attempting to alleviate the stress and declines to these vulnerable populations worldwide. It has become necessary and significantly more common in recent decades for the composition, distribution, and especially the diversity of reptiles and amphibians to be studied to assess deviations from expected patterns, evaluate ecological state, estimate causes of decline, and to inform species and habitat conservation priorities and tactics (Leyte-Manrique et al. 2019, Russell et al. 2002, Urbina-Cardona 2008).

Detection studies and diversity assessment have increasingly become a critical component of studies assessing herpetofauna in particular – their status, abundance and distribution, habitats, and priorities in conservation. Such studies largely employ methods such as Simpson's Diversity Index and the newer-developed Shannon-Weiner index to describe an ecosystem's diversity with a value instead of measuring solely relative abundance of species. Studies also utilize other species measures as trends over time such as abundance, dominance, rarity, evenness, and richness in order to inform ecologists on the composition, state, health, and conservation threat of specific herpetological communities (Briones Jr. 2017, Kumar et al. 2022, Smith and Berry 2010, Wilson and Gownaris 2022). Considering that herpetofauna species are notoriously difficult and inconsistent to detect, challenging to monitor, knowledge and surveys in

many of their critical regions remain incomplete, their habitats have been largely degraded or fragmented to varying extents, species have differed levels of threat or decline, little information is available on how factors affect their species assemblages, and raw counts or similar metrics are highly susceptible to bias, these two groups of species are and have been a major suitable target for the newer-developed detection, management, and conservation strategies utilizing species diversity indices and other related species trend measures (Behangana et al. 2020, Kumar et al. 2022, Kusamba 2019, Smith and Berry 2010).

As mentioned, herpetofauna across the board experience heightened levels of threat and vulnerability to a wide variety of factors causing their global declines, namely habitat disturbance and fragmentation due to anthropogenic, climate, or other origins (Ruthven et al. 2002, USGS 2022, Whitfield Gibbons et al. 2000). Diversity index studies taking into account abundances, evenness, distribution, dominance, and other factors allow spatial status and occurrence assessment, comparative studies among and between numerous fragmented habitats, species response to environmental disturbances or degradations, contributions of specific habitats to overall populations and species continuation, species assemblages in differing landscapes, species dominance or rarity, and identification of species, habitats, and populations most at risk, therefore contributing to overall knowledge and prioritization of those when it comes to allocating conservation attention, funding, and efforts (Ghosh, Kusamba 2019, Laurencio and Fitz, Leyte-Manrique et al. Mortiz). A major 2019 study on herpetofauna in the Democratic Republic of the Congo utilized long term herpetological surveys including species occurrence and relevant diversity indices, including Shannon-Weiner index, along with richness, rarity, presence of threatened species, and endemism in order to direct conservation awareness and governmental and organizational efforts to appropriate habitats. This ultimately led to the

identification and creation of ten sites as the proposed new Protected Areas for reaching the government's national conservation targets necessary for conserving the rich biodiversity of the country (Kusamba 2019). Another major study in 2019 and one in 2002 concluded the significant importance of herpetological diversity (both using Shannon-Weiner diversity index), abundance, and taxonomic distinctness of environments with different levels of disturbance in the identification and development of efficient conservation routes and strategies for reptiles and amphibians in Salvatierra, Guanajuato, Mexico and coastal South Carolina, respectively (Leyte-Manrique et al. 2019, Russell et al. 2002). It is clear that species diversity and its associated measures are principal to informing herpetological conservation on the basis of their largest threats – habitat degradation, fragmentation, lack of knowledge and historical surveys, and anthropogenic disturbance, among many others.

The in-situ field portion of this study, occurring over a period of four years in Northern Mississippi, is a herpetological survey which then had mathematical indices and measures incorporated in an attempt to analyze community composition, species diversity, and other species measures as trends over time. The goal of this study was to ultimately report herpetofauna species found in the region, their presence and detection, variation, diversity, abundance, dominance and rarity, richness, and other measures throughout the four separate survey trips, showing trends over the periods of time in order to better inform the timing and methods for continuation and amplification of the survey in future years. It also emphasizes deviations from expectations for these parameters, giving insight into the state of the herpetofauna community in the study region, creating a baseline for comparison in continuation of the study, and noting areas of conservational concern.

Methods

The goal of each trip within the overall study was a survey of the herpetological species in the area using varying in-situ detection methods and later ex-situ data analytical methods.

In-situ Fieldwork Methods

In-situ methods occurred in remote upland Northern Mississippi over the course of four years: 2018, 2019, 2021, and 2022. The 2018 trips and data collection occurred in mid-April, again in late May and early June, and a third time in later June. The 2019 trip and data collection occurred in late March and early April. The 2021 and 2022 trips and data collection both occurred in early October. The habitat of the study region is mixed prairie and field of retired farmland, hardwood forest with both flat and sloped landscape, and scattered standing water and streams with riparian habitat.

Herpetological surveying was carried out utilizing multiple various methods of detection targeting different types of reptiles and amphibians. Coverboards, ranging from 3x5 feet to approximately 6x10 feet and made of either corrugated tin or plywood, were arranged within the study area at varying elevations, slopes, landscapes, and amount of canopy cover (boards utilized ranged from no to full cover). These boards were placed mainly away from larger sources of water, being positioned within either leaf litter and soil forest floor or grassy field. This type of detection method targets both reptiles and amphibians due to their shared characteristic of having strong, distinct hygothermal microclimate preferences (Grant et al. 2001). The habitat of herpetofauna across the board is wholly based upon the heat and moisture, or hygothermal, nature of the substrate and air around them, with each species occupying a very specific microclimate exhibiting those particular levels to them (Grant et al. 2001). Coverboards offer the manmade creation of a variety of microclimates placed where they may not have existed naturally, therefore valuable to designing a successful study region for a survey of this type.

However, coverboards in such surveys target reptiles more in particular, due to the heat absorbing nature of the boards' large surface area, attracting these more highly ectothermic, less permeable-skinned, and drier climate species more resistant to desiccation. Reptiles are more likely to move under these boards and obtain heat directly from the board above them, particularly when the ground below is drier, than amphibians are expected to (Grant et al. 2001). However, this method does not entirely exclude amphibian species, as some of the larger, thicker wood coverboards harbored a decent amount of moisture under them, particularly if the board was atop soil and especially if it was under a larger amount of canopy cover causing more shading and slightly cooler temperatures. This aspect allows for species more permeable-skinned and more vulnerable to desiccation to move to them for a heat source without too dry of a substrate. Coverboards were checked daily in each trip of the study by approaching and carefully lifting each board away from the individual or group of individuals present as a safety precaution, and then quickly assessing the area under the board and trapping any animals found. Carpets were used similarly, but only two of these were placed in the study region. These fabric-like mats were approximately 12x12 feet each, both positioned on a hillier landscape directly atop soil substrate and under approximately 30% canopy cover. These carpets target both types of herpetofauna, rapidly absorbing more heat due to their material and higher sun exposure, as well as harboring larger amounts of moisture between mat and substrate after a precipitation event. The by-hand trapping of the study made up the largest portion of detection methods used, detecting and trapping herpetofauna by physically displacing parts of or objects in the landscape acting as natural "coverboards" in an attempt to reveal animals below. These included natural parts of the animals' habitats, mainly "dead and down" tree branches and trunks both newly downed or decomposing, leaf litter, rocks, and scrap wood, metal, and other materials not

necessarily placed to carry out the function of a coverboard (present mainly near the garages throughout the study area). This detection method was carried out throughout the day on each day of each trip, starting early in the morning and continuing until evening as the region was traveled and other forms of traps were checked. This method targeted both kinds of herpetofauna, as both utilize such objects and parts of their natural landscape and habitat to obtain or conduct heat and moisture. A reptile may be present under a rock in direct sunlight in order to conduct heat, while an amphibian may be present when a moist, decomposing log is overturned. This detection and trapping method was carried out in all types of habitat across the study region, not being limited to drier or wetter conditions or a particular type of substrate, canopy cover, etc. Visual detections also included a significant portion of detections which also led to hand-trapping. This type of detection method, unlike the last, involved hand trapping herpetofauna after purely visual detection only, without the physical disturbance of any piece of the environment involved. This method is one which does favor reptilian over amphibian species. Reptilian species, being less resistant to desiccation and therefore less reliant on moisture, shade, or cooler temperatures, use the behavior of basking frequently in order to maximize the amount of heat conducted or obtained from their environment. Basking is the main behavior leading to visual encounters and detection without disturbing or physically altering the environment, as individuals must be exposed in order to carry out this heat exchange with their habitat. It is not common for amphibious species to bask, but is a behavior shared by almost every type of reptile. Finally, ProMar 250 hoop funnel nets were employed in two major bodies of water within the study region, both standing, freshwater ponds. Two traps were placed and secured within the smaller pond while the number of traps in the larger body of water ranged from 3 to 6 traps. These were baited with canned cat food, replaced each day as the traps were

pulled in and checked daily for each day of each trip. This detection method clearly targets both aquatic reptilian and amphibian species rather than the strictly terrestrial targets of the other detection methods.

As each detection method was checked or carried out, any herpetofauna found or caught were placed into appropriate containers, either plastic jars or fabric snake bags based on species and size. Turtles were placed in snake bags, while smaller reptiles and amphibians were mainly contained within the jars. While still in-situ, data based on the environment was collected and recorded, along with a number identifying the animal with its data sheet. For each capture, while still in-situ, field identification was carried out based on initial impressions and prior herpetological knowledge, and then date, time of capture, latitude and longitude of capture site, substrate and habitat type, and measures of air, substrate, and ground (if different) temperatures in degrees Celsius, percent relative humidity, windspeed, and canopy cover percentage were recorded with varying equipment, along with notes of any distinguishing characteristics of the individual or any comments on the capture. Each capture site was also marked in the GPS units as a specific numbered waypoint corresponding with the animal and data sheet. Animals in appropriate containers were secured by members of the group and were taken from the field to, in the case of this study, the cabin in the study region, for processing and further ex-situ data collection.

Once present at the processing site, animals were matched with their data sheets from the field by number, and then handled and assessed further one by one, prioritized based on time since capture, time away from moisture or heat sources, and apparent condition of the animal. Species of each animal was confirmed by literature and recorded, and weight, plastron and carapace length and width for turtles, snout to vent length for other species, sex, and age

category were also recorded with the animal's ID. Individuals large enough (such as turtles or larger snakes) were assessed for presence of a PIT tag, and, if not a recapture or previously PIT tagged, were then cleaned with antiseptic pads, injected with the tag, and the wound was covered with Liquid Skin agent to promote healing. After all data was recorded, animals were returned to exact capture site found by marked GPS points and released.

This process of in-situ capture, in-situ data collection, transportation, ex-situ data collection, and release to wild was repeated once or twice daily for each day of each trip, besides the trip during the summer months, in which a different method of detection focusing on salamanders created an anomaly for this season in the entire study overall, as noted in the results.

Data Analytical Methods

Ex-situ analysis of all field-collected data was the second major aspect of this study. Data analytical methods following the completion of all of the trips analyzed multiple aspects of the results of the four-year herpetological survey.

All data was input into Microsoft Excel separated by trip or year, as this is how the data sheets themselves were initially organized. Capture data was then separated by "season," determined by month. Records from the end of March and beginning to mid-April were deemed as "spring," the end of May and early to late June were deemed "summer," and early October was deemed "fall." This separation of year or trip and season designation would allow for the diversity and other measures as trends to be compared between the two different types of time period. Data input into Excel for each capture was pulled directly from the in-situ data sheets and included animal ID number, species designation, date of capture, time of capture, air temperature in degrees Celsius at the time of capture, and substrate temperature in degrees Celsius at the time of capture.

After this initial data was input, total numbers, or abundance (N or n), of the two broadest types of herpetofauna (reptiles and amphibians) and each type of herpetofauna species (salamanders, frogs, turtles, lizards, etc.) were counted overall for all four trips combined and then separated by year or trip and then again by season, giving a total number of individuals captured for each year, for each season, and overall for the four year period. Pie charts were created from these totals, showing proportions of each species group relative to one another from the study overall, by year, and by season, and again showing proportions of each of the two broad types of herpetofauna relative to one another from the study overall, by year, and by season.

Proportion (p_i) of each species per year or trip was calculated by dividing the number of individuals in that group caught in each year by the total number of individuals overall caught in that year, as in the example below with skinks.

$$\text{Skinks as proportion } (p_i \text{ skinks}) = \frac{\text{number of individual skinks captured in the time period}}{\text{total number of herpetofauna individuals captured in the same time period}}$$

This was done for each year or trip and then for each season. These proportions are all fractions of 1 so that the sum of all species detected in a particular time frame would sum to 1. The maximum proportion a species can exhibit during a particular time frame is 1, if that was the only species detected. Pie charts showing species proportion for each trip were created. Proportions of each broader type of herpetofauna (reptiles and amphibians) were calculated in the same manner, as below, for each year or trip and for each season.

$$\text{Reptiles as proportion } (p_i \text{ all reptiles}) = \frac{\text{number of total reptiles captured in the time period}}{\text{total number of herpetofauna individuals captured in the same time period}}$$

Pie charts were also created with these proportions as well as line graphs plotting proportion over time. Two lines were plotted, one for reptiles and one for amphibians, with year or season on the

x-axis. These proportions had the same range of 0-1 with all species detected in a time frame summing to 1.

Species richness was calculated for each trip or year and then for each season from the number of species that was detected during that time period. For instance, if in 2018 two snakes, one skink, and five turtles were caught, but no frogs or salamanders, the species richness for the 2018 trip would equal 3. Species richness along with abundance were recorded for use in the next equations to find species diversity.

Species diversity calculation was the next major assessment performed. Diversity indices are a mathematical measure of species diversity and the Shannon-Weiner Diversity Index equation was one utilized in this study. This index is one of multiple species diversity indices employed in the ecology field, yet it is particularly beneficial when estimating the diversity of species, or the heterogeneity, within an ecological community (Martensson 2016). The Shannon Index takes into account both species richness and relative species abundance, as mentioned, and better reflects overall community biodiversity than other ecological diversity indices (Briones Jr. 2017). The Shannon Index value, or Shannon value, increases with diversity. The Shannon Index is also related to the concept of uncertainty. For example, if a community has very low species diversity, using the Shannon Index, the identity of an organism chosen by random can be determined with higher certainty or confirmed with a lower amount of uncertainty. If a community has a high diversity and an organism is chosen by random, there is greater uncertainty surrounding the identity of the species chosen, in other words, there is either low certainty or high uncertainty in this case (Wilson and Gownaris 2022). The Shannon Index was created on the basis of describing the “entropy,” or “disorder and uncertainty” of individual species in a community (Ekberzade 2022, Kaur 2020). The higher the uncertainty of the species

identity of a chosen organism from the community, the higher the species diversity is implied in that community (Kaur 2020). Due to the nature of this equation, either a large number of species detected in the community, or a more even uniformity between the abundances of individual species detected in the community (species evenness) will increase the diversity index of that community. The equation for the Shannon-Weiner Index follows.

$$H = - \sum_{i=1}^S p_i * \ln p_i$$

For these reasons, the Shannon-Weiner Index was an appropriate calculation of species diversity in this study. Using species identification, overall species abundance (N), and species richness (S) for each time frame or community, the Shannon-Weiner value (H) is formulated by first calculating the proportion (p_i) of each species, as calculated earlier in the study methods – dividing the number of individuals in the species detected in the time frame (year or season) by the total number of individuals detected in the time frame. For each species, that calculated proportion is then multiplied by the natural log of that proportion ($\ln p_i$). All of the values calculated for each of those species during the time frame are then summed and finally are multiplied by -1, giving the final Shannon Index value indicating degree of diversity using a degree of certainty (Wilson and Gownaris 2022). The Shannon-Weiner Diversity Index of values ranges from 0 to $\ln N$, with 0 being the lowest possible and $\ln N$, or natural log of the total number of species detected, being the highest possible species diversity in a given community considering that this value has a positive correlation with community biodiversity (Wilson and Gownaris 2022). If the individuals detected in a community were to all belong to a different species, the diversity index would reach its maximum, while if each individual detected were to belong to the same species, this index would be at its minimum value of 0 (Rain 2022, Wilson

and Gownaris 2022). Any variations of amounts of individuals per species detected between those two extremes will return values between 0 and $\ln N$, with most normal diversity values falling between 1.5 and 3.5 (University of Idaho 2009). In order to carry out these diversity calculations in the study, the already established columns of species groups, their abundance per trip year or season, the overall species abundance per trip or season, and the overall species richness per trip or season were utilized. In the manner described above, proportions were first calculated for each species group abundance in the 2018 trip and continuing through all trip years and then each of the three seasons. A Shannon-Weiner Index value was calculated for each of these time frames, acting as separate “communities” separated by time rather than physical location to rather assess diversity trends over a temporal rather than geological scale. After these values were calculated and recorded accounting for all species groups of herpetofauna detected, each time frame was then split into two separate charts of data, one with only the amphibian species and one with only the reptilian species. The Shannon-Weiner Index was then calculated in the same manner, resulting in a separate amphibian-only and reptile-only diversity value for each time frame of trip and season.

In addition to the Shannon-Weiner Index of diversity, another frequently used diversity index in the ecological world was also performed on the data collected from this study. The Simpson’s Diversity Index exhibits several differences from the Shannon-Weiner Index, but is also valuable to include in ecological community surveys (Briones Jr. 2017, Wilson and Gownaris 2022). This index is based upon a measure of probability – the less diversity in a community, the greater the probability that two randomly selected individuals from that community will be of the same species (Wilson and Gownaris 2022). In the absence of diversity,

if there were only one species present, the probability that the two individuals selected will be of the same species is 1. The equation of Simpson's Index follows.

$$D = \sum_{i=1}^S \left(\frac{n_i}{N} \right)^2$$

Using species identification, overall species abundance (N), and species richness (S) for each time frame or community, the Simpson's Diversity Index value (D) is formulated by first calculating the proportion of each species, as calculated earlier in the study methods and in the Shannon-Weiner Index – dividing the number of individuals in the species detected in the time frame (year or season, n) by the total number of individuals detected in the time frame (N). For each species, that calculated proportion is then squared. All of the values calculated for each of those species during the time frame are then summed. The value of Simpson's diversity ranges from 0 to 1, but in a different way than the Shannon-Weiner Index. 0 in this case, due to the nature of the equation, represents infinite diversity and 1 represents no diversity, so the larger the initial value of D , the lower the diversity. For this reason, the Simpson's Index value is often actually represented by $1-D$, showing instead the diversity of the community on a scale from 0 to 1, 1 representing maximum diversity (Wilson and Gownaris 2022). The initial D value was, for that reason, then subtracted from 1 to give the final Simpson's Diversity Index value for each time period (trip or season) in the study. A Simpson's Index value was also calculated for each of the time frames acting as separate temporal communities rather than community based on physical location to similarly assess diversity trends over a temporal rather than geological scale. The Simpson's Index was also calculated in the same manner on the data split by broader amphibian and reptilian species, resulting in separate amphibian-only and reptile-only diversity values for each time frame of trip and season. Due to its nature, Simpson's Index measures

diversity using purely the relative abundance of each species in the community with each value showing the dominance or non-dominance of that particular species (Briones Jr. 2017). This index emphasizes more the common or dominant species in an index, while the Shannon-Weiner Index, as mentioned, better emphasizes and gives more weight to rarer species detected within a community (Smith and Berry 2010). Unlike the Shannon-Weiner Index, Simpson's ignores species richness and equability as a factor into community biodiversity (Briones Jr. 2017). However, both indices are often employed together on such survey studies in order to assess community diversity using the different parameters and emphases of both measurements, and comparison between them is valuable to drawing conclusions on true community diversity as well (Beals et al. 1999, Briones Jr. 2017, Martensson 2016, Mohammed 2021). For this reason, both were calculated on the same data and their diversity values compared over the study timeframes.

Species evenness was another parameter measured in this study in order to be analyzed as a trend over time. Species evenness refers to how close in number each species, or in the case of this study, each broader species group, in a community is. It refers to increased uniformity between the abundances of each species group detected to one another, and is another parameter used in ecological community studies to give a rough idea of diversity and to measure species dominance or rarity, the somewhat opposing force to diversity, in a community (Wilson and Gownaris 2022). Species evenness is calculated for a community or, in the case of this study, for each time frame of trip or season, by dividing the Shannon-Weiner diversity value for that period (H) by the natural log of S , the overall species richness for that period, as below (Wilson and Gownaris 2022).

$$J = H/\ln(S)$$

This index was performed on all species groups for each time frame – trip and season. The range of values for species evenness is 0 to 1, with 1 indicating complete equability or uniformity between species detected in that community (Rain 2022).

Finally, temperature as a parameter was assessed for this study, based on the strong relationship between herpetofauna species behavior and the temperatures of their environments. These averages would be used to compare to the measures described prior, such as abundance, diversity, richness, evenness, dominance and rarity, etc. that were calculated from this survey, and also in comparing findings with similar past studies from literature. Average temperature in degrees Celsius of each year or trip was calculated by adding each air temperature recorded at every capture during each trip or season, and then dividing that number by the amount of air temperature recordings taken. The same was done for each air temperature reading from each season, giving an average temperature for each season as well. Both the air and substrate temperatures, if recorded, for amphibians and reptiles were separated. In the same manner, an average air temperature, then an average substrate temperature, then an average overall temperature for amphibian species was calculated. An average air, then substrate, then overall temperature was calculated for reptilian species as well. The highest and lowest air and substrate temperature at a detection was noted for amphibians and then for reptiles for analysis uses. Substrate temperature was added at this point to assess another facet differentiating amphibians and reptiles: how they behave in response to temperature of the environment directly around them (the substrate they are conducting heat from).

All of these calculations and recorded values were then placed into pie charts, line graphs, or tables where appropriate, and analyzed in context, comparing the expectations and

predictions for the survey outcomes based upon cited literature and personal herpetological knowledge to what was actually observed over the four-year period.

Results

In-situ Results

Over the course of this four-year study period, a total of 86 herpetofauna individuals were captured, detecting 26 distinct species, or an overall species richness of 26. Reptilian species encompassed 18 of the 26 distinct species. Amphibian species accounted for the remaining 8 distinct species detected. The reptilian species detected in the community included black-masked racer (*Coluber constrictor latrunculus*), green anole (*Anolis carolinensis*), brown anole (*Anolis sangrei*), corn snake (*Pantherophis guttatus*), ring-necked snake (*Diadophis punctatus*), southeastern 5-lined skink (*Plestiodon inexpectatus*), little brown or ground skink (*Scincella lateralis*), eastern fence lizard (*Sceloporus undulatus*), southeastern crowned snake (*Tantilla coronata*), red-bellied snake (*Storeria occipitomaculata*), eastern worm snake (*Carphophis amoenus*), king snake (*Lampropeltis getula nigrita*), 3-toed box turtle (*Terrapene carolina triunguis*), plain-bellied water snake (*Nerodia erythrogaster*), common mud turtle (*Kinosternon subrubrum*), smooth earth snake (*Virginia valeriae*), eastern box turtle (*Terrapene carolina carolina*), and common 5-lined skink (*Plestiodon fasciatus*). Amphibian species detected in the community included Mississippi slimy salamander (*Plethodon mississippi*), southern cricket frog (*Acris gryllus*), American toad (*Anaxyrus americanus*), southern toad (*Anaxyrus terrestris*), eastern cricket frog (*Acris crepitans*), southern 2-lined salamander (*Eurycea cirrigera*), pickerel frog (*Lithobates palustris*), and red-spotted newt (*Notophthalmus viridescens*).

Data Analytical Results

Ex-situ calculations and formulations led to intriguing findings based upon the in-situ detections and data collection. Below are the results of the aforementioned ex-situ methods carried out separately from the fieldwork.

Firstly, as mentioned, 86 individuals were detected over the four trips, with the amount of reptiles (50) detected larger than the number of amphibians detected (36). This resulting ratio of

reptiles to amphibians detected overall in the study was 1.39:1. Overall species abundance per year was calculated and it was found that the 2018 trip resulted in the most herpetofauna detections, having the highest abundance of organisms detected. The 2019 trip resulted in the least amount of herpetofauna detections, having the smallest abundance of organisms detected by a significant amount (Table 1a). It was also found that relatively close numbers of organisms were found in spring and fall, but more were found in the fall than spring seasons (Table 1b).

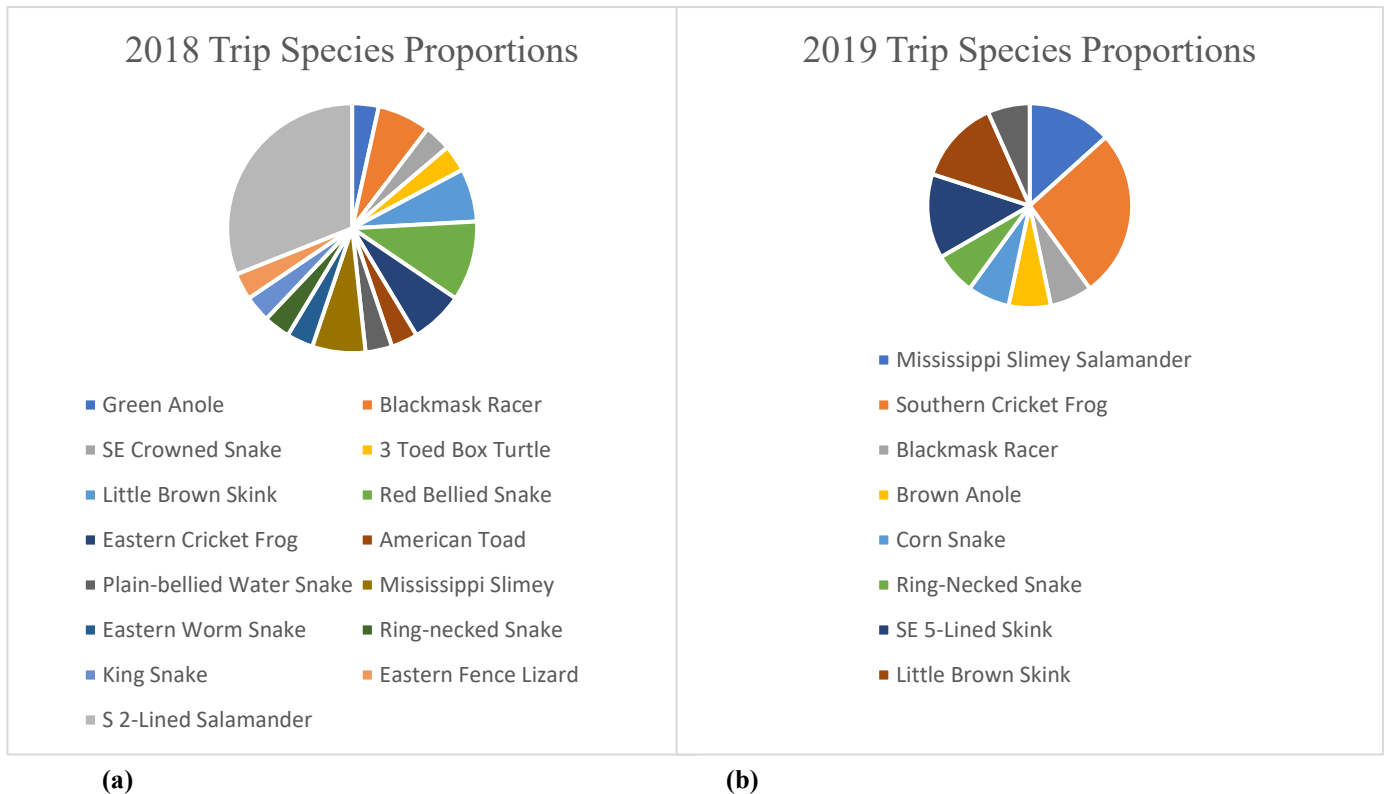
Overall Species Abundance per Year/Trip		Overall Species Abundance per Season	
2018	29	Spring	35
2019	15	Summer	9
2021	21	Fall	42
2022	21		

Table 1a-b. Overall species abundance tables. The overall abundance or number of individuals detected per timeframe, year or trip (a) and season (b) are arranged in tables.

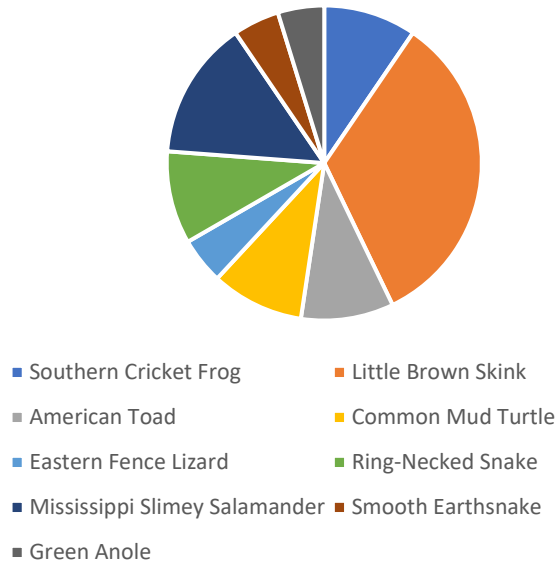
Proportions then calculated were translated into pie charts illustrating the portion of detections that first, each species made up, and second, both broader types of herpetofauna made up, over both the time frames. This also assessed species dominance or rarity across the timeframes. It was found that the species of most detection was the southern two-lined salamander in 2018, the southern cricket frog in 2019, the little brown skink in 2021, and the little brown skink again in 2022 (Fig. 1a-d). Similarly, the proportions and dominance of each species was assessed by season as well. The species of most detection in the spring season were equally the Mississippi slimy salamander, southern cricket frog, and little brown skink, while the species of least detection included the brown anole, the corn snake, the green anole, the southeastern crowned snake, the three-toed box turtle, the American toad, the plain-bellied watersnake, the eastern wormsake, and the king snake. The species of most detection in the fall season was overwhelmingly the little brown skink, while the species of least detection included the eastern fence lizard, the smooth earthsnake, the southern two-lined salamander, the eastern

box turtle, the common five-lined skink, the pickerel frog, and the red-spotted newt (Fig. 1e-f).

Overall, the species of highest detection were the little brown skink followed by the southern two-lined salamander and then the Mississippi slimy salamander, making these the dominant species detected across the entire study (Fig. 1g).

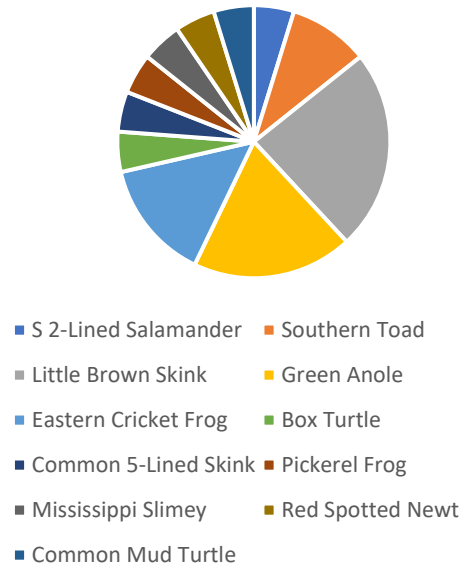


2021 Trip Species Proportions



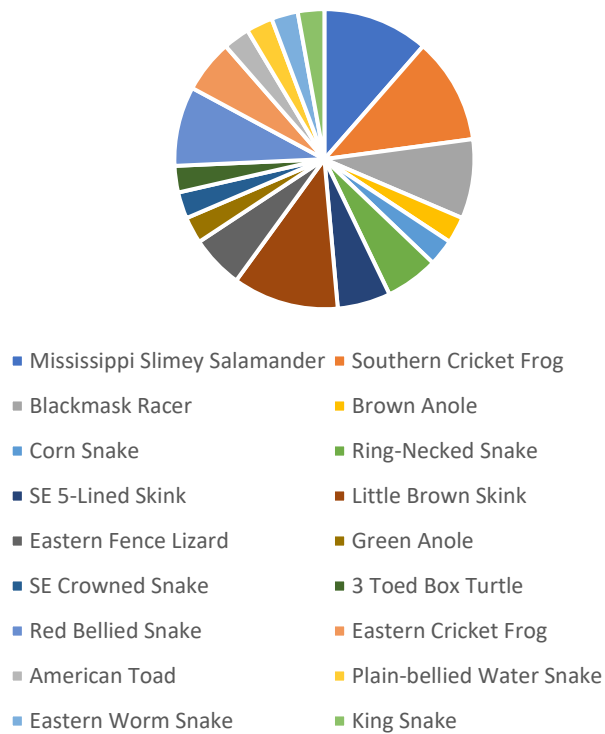
(c)

2022 Trip Species Proportions



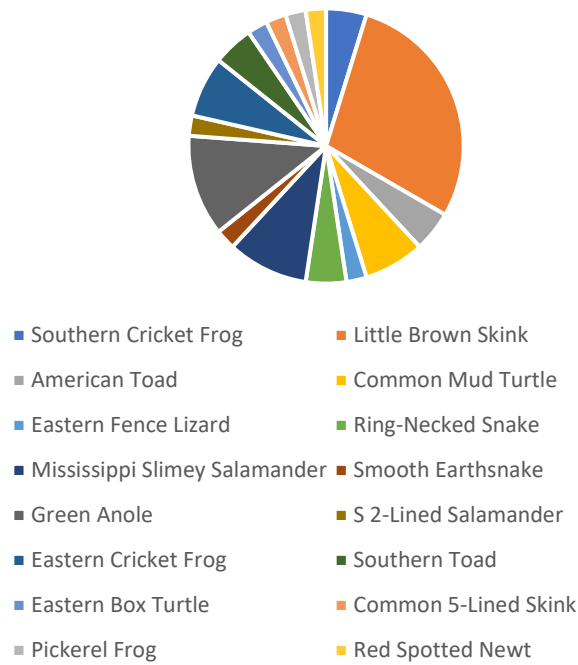
(d)

Spring Season Species Proportions



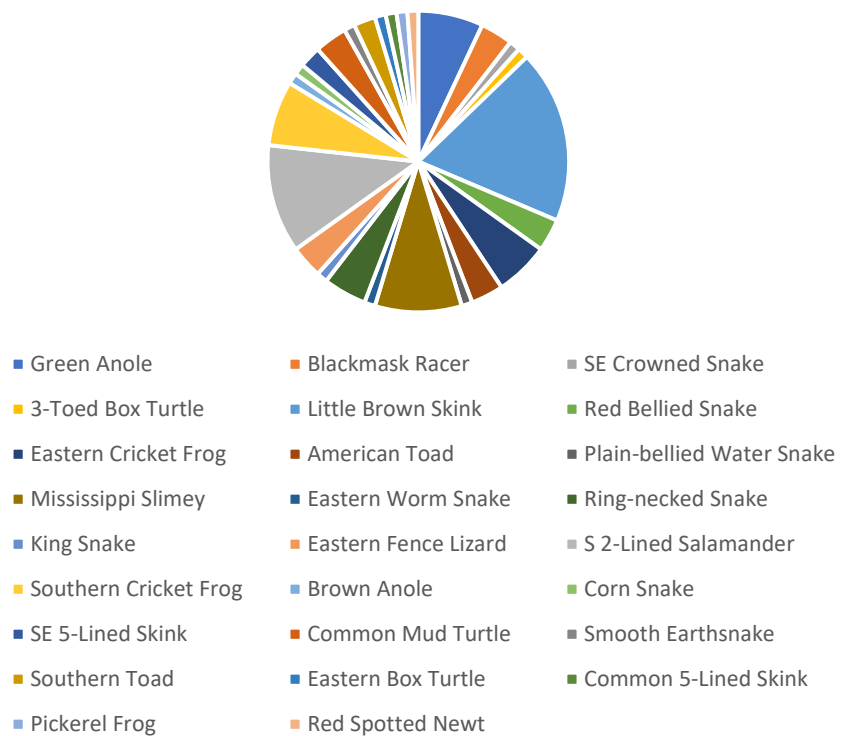
(e)

Fall Season Species Proportions



(f)

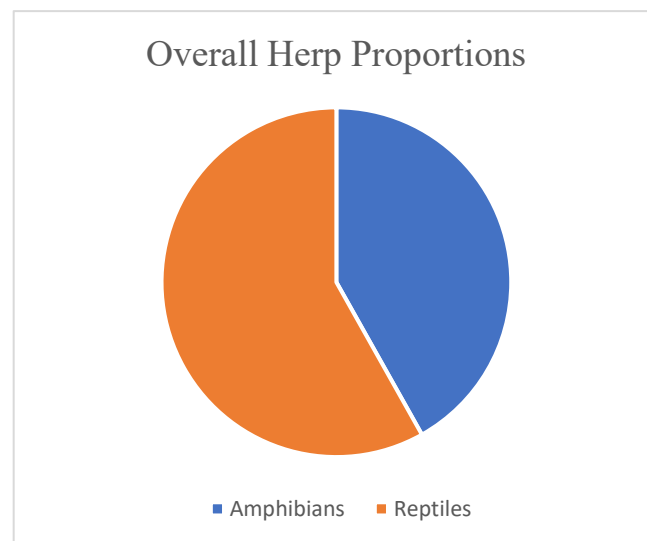
Overall Species Proportions



(g)

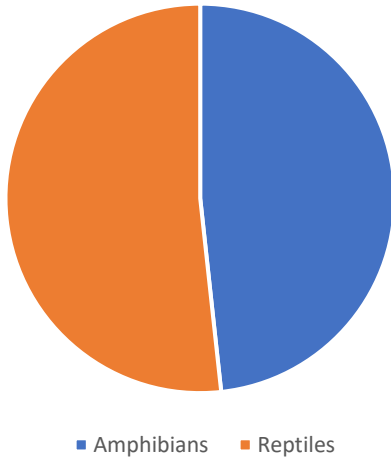
Figure 1a-g. Proportion pie charts. The proportions of species detected per year or trip (a-d), season (e-f) and across the trip overall (g) arranged in pie charts.

The proportions and dominance for these same time periods, but instead considering more broadly the two types of herpetofauna, reptiles and amphibians, were also placed into pie charts. When assessed for overall survey proportion, again, reptiles were the more detected and dominant type of herpetofauna, and this proved true in all but one other instance. In the 2018, 2019, 2021, and 2022 trips, reptilian species groups were the more detected and dominant herpetofauna detected. However, in 2018, the proportions of reptiles and amphibians was almost equal (Fig. 2b). This year showed the greatest amount of evenness in detection found between the two types of organisms out of all time frames. Overall survey, 2019, and 2022 showed very similar relative proportions (Fig. 2a, c, and e), while 2021 showed the most reptile-dominated trip when considering year (Fig. 2d). Proportion of detections in the spring season showed the most reptile-dominant season out of the three season time frames (Fig. 3a), while summer obviously exhibited the most amphibian-dominated season (Fig 3b). The amphibian detection proportion was higher relative to the reptilian detection in the fall than in the spring.



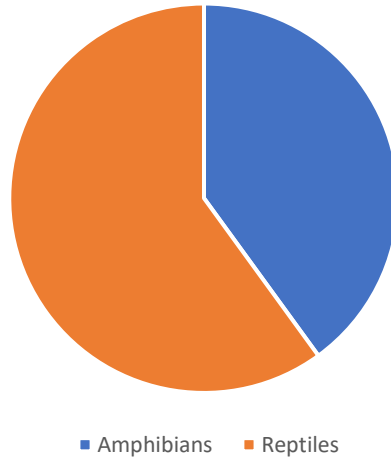
(a)

2018 Trip Herp Proportions



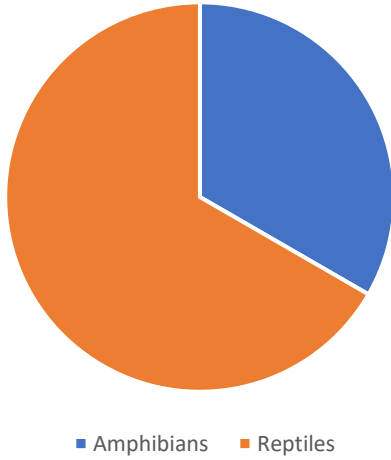
(b)

2019 Trip Herp Proportions



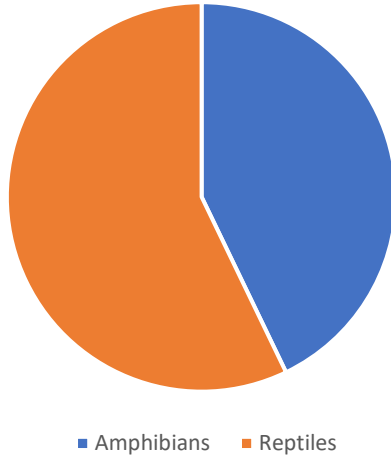
(c)

2021 Trip Herp Proportions



(d)

2022 Trip Herp Proportions



(e)

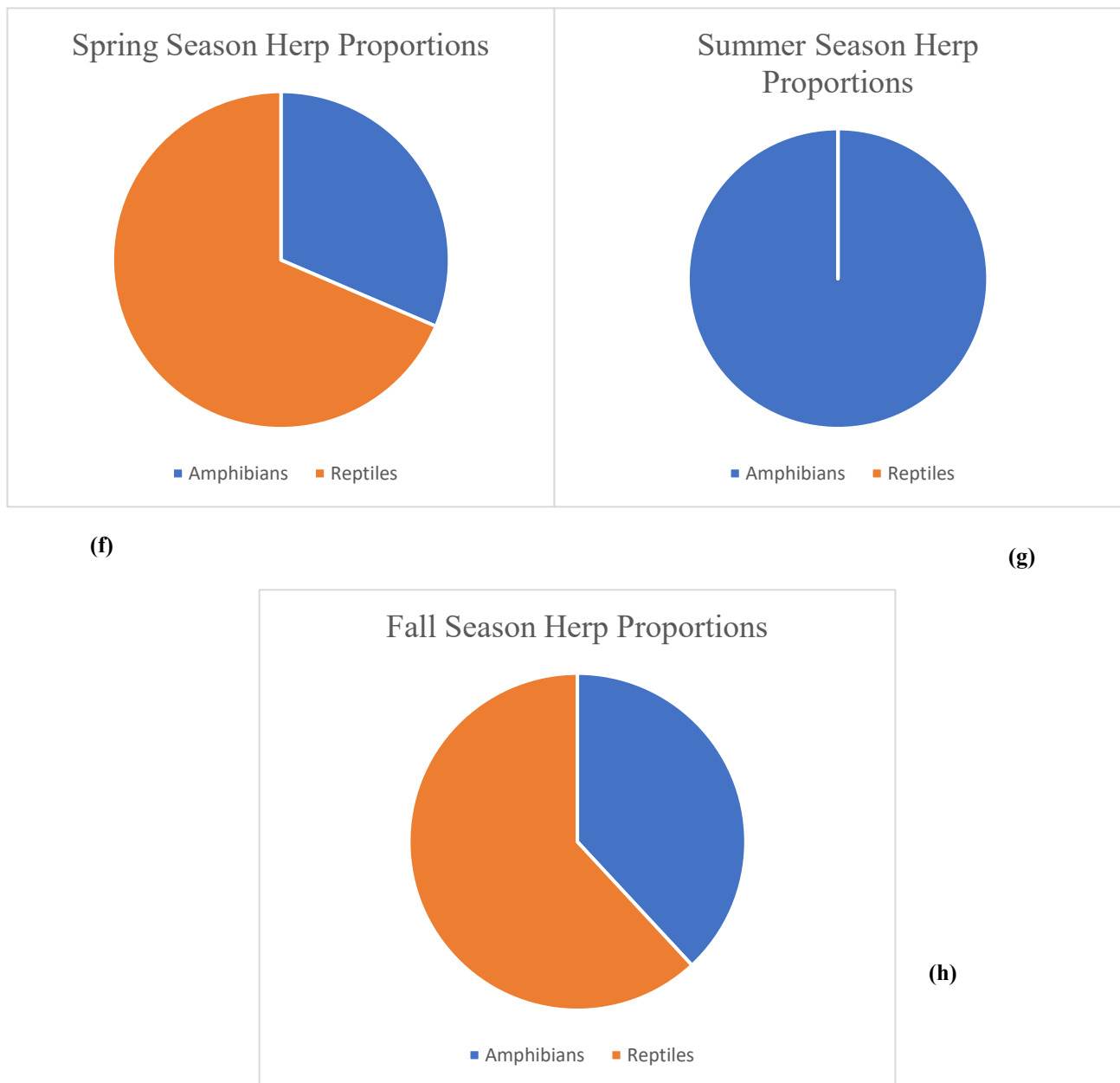


Figure 2a-h. Proportion pie charts. Proportion pie charts. The proportions of broader type of herpetofauna detected overall across the study (a), per year or trip (b-e), and season (f-h) arranged in pie charts.

As with the pie charts of proportions, line graphs of proportion of reptiles and amphibians over year time frames and season time frames similarly show the constant dominance of reptilian species detection over amphibian, but the years where these proportions are closer are also observed (Fig. 3a-b).

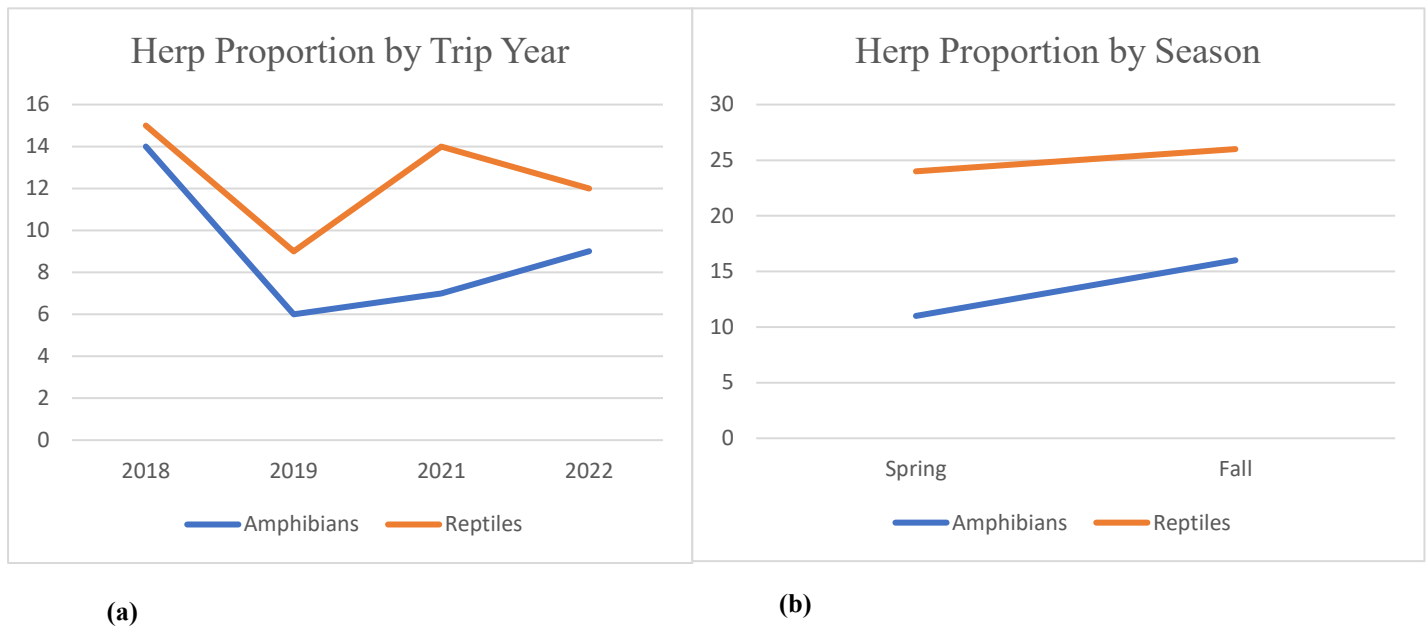


Figure 3a-b. Proportion graphs. The proportion of broader type of herpetofauna are arranged in line graphs over year or trip (a) and season, excluding the summer anomaly (b).

Species richness calculations by year and season were placed in tables (Table 2a-b). The time frames with less or more species richness are apparent. 2019 and 2021 had the equally lower amount of richness, while 2018 and 2022 had higher amounts of distinct species detected (Table 2a). The drastic drop anomaly in species richness is observed in the summer, where spring and fall showed detection of 18 and 16 distinct species, respectively (Table 2b).

Species Richness by Year/Trip		Species Richness by Season	
2018	15	Spring	18
2019	9	Summer	1
2021	9		
2022	11	Fall	16
(a)		(b)	

Table 2a-b. Species richness tables. The species richness, or number of distinct species, detected per timeframe, year or trip (a) and season (b) are arranged in tables.

Shannon-Weiner Diversity Index values were also arranged into tables. It was found that the highest amount of diversity using this index occurred in the 2018 trip and also in the spring season (Table 3a and b, respectively). The lowest amount of diversity using this index was found

in the 2021 trip and in the summer anomaly season, however, the diversity using this index of fall was only slightly lower than that found in spring (Table 3a and b, respectively).

SW Index Values by Trip/Year SW Index Values by Season

2018	2.370	Spring	2.741
2019	2.063	Summer	0
2021	1.976	Fall	2.420
2022	2.180		
(a)		(b)	

Table 3a-b. Shannon-Weiner diversity index tables. The calculated Shannon-Weiner Diversity Indices for each trip or year (a) and season (b) are arranged in tables.

Once split into broader type of herpetofauna, it was found that amphibian species had lower Shannon-Weiner diversity values in almost every trip or season time period except for the 2022 trip, where amphibian diversity using this index exceeded reptilian (Table 4a and b). In the 2019 trip, the difference in type of herpetofauna diversity using this index was greatest, as reptilian diversity value was almost triple that of the amphibians (Table 4a and b). The highest diversity calculated using this index was found in the reptiles in 2018 (Table 4b).

SW Index Values for Amphibians by Trip/Year SW Index Values for Reptiles by Trip/Year

2018	1.028	2018	2.307
2019	0.667	2019	1.891
2021	1.079	2021	1.466
2022	1.676	2022	1.351
(a)		(b)	

Table 4a-b. Shannon-Weiner diversity index tables. The calculated Shannon-Weiner Diversity Indices for amphibians per year or trip (a) and for reptiles per year or trip (b) are arranged into tables.

As for diversity between types of herpetofauna based on season, reptiles showed much greater diversity using this index in the spring, however, amphibians showed the larger amount of diversity using this index in the fall season (Table 5a-b).

SW Index Values for Amphibians by Season

Spring	1.264
Summer	0
Fall	1.963
(a)	

SW Index Values for Reptiles by Season

Spring	2.504
Summer	0
Fall	1.617
(b)	

Table 5a-b. Shannon-Weiner diversity index tables. The calculated Shannon-Weiner Diversity Indices for amphibians per season (a) and for reptiles per season(b) are arranged into tables.

Simpson's Diversity Index values were also arranged into tables. It was found that the highest amount of diversity using this index also similarly occurred in the 2018 trip and also in the spring season (Table 4a and b, respectively). The lowest amount of diversity using this index was found in the 2021 trip and in the summer anomaly season, however, the diversity using this index of fall was also lower than that found in spring (Table 4a and b, respectively). Compared to the Shannon-Weiner indices, these values show the same trends but much closer values relative to each other.

Simpson's Index Values by Trip/Year

2018	0.864
2019	0.853
2021	0.826
2022	0.862
(a)	

Simpson's Index Values by Season

Spring	0.926
Summer	0
Fall	0.872
(b)	

Table 4a-b. Simpson's diversity index tables. The calculated Simpson's Diversity Indices for each trip or year (a) and season (b) are arranged in tables.

Once split into broader type of herpetofauna, it was similarly found that amphibian species had lower Simpson's diversity values in almost every trip or season time period except for the 2022 trip, where amphibian diversity using this index exceeded reptilian (Table 5a and b). In the 2019 trip, the difference in type of herpetofauna diversity using this index was again the greatest, as reptilian diversity value was about double that of the amphibians using this diversity index

(Table 5a and b). The highest diversity calculated using this index was also found in the reptiles in 2018 (Table 5b).

Simpson's Index Values for Amphibians by Trip/Year

2018	0.542
2019	0.444
2021	0.652
2022	0.791

(a)

Simpson's Index Values for Reptiles by Trip/Year

2018	0.889
2019	0.840
2021	0.694
2022	0.695

(b)

Table 5a-b. Simpson's diversity index tables. The calculated Simpson's Diversity Indices for amphibians per year or trip (a) and for reptiles per year or trip (b) are arranged into tables.

As for diversity between types of herpetofauna based on season, reptiles again showed much greater diversity using this index in the spring, and again, amphibians showed the larger amount of diversity using this index in the fall season (Table 6a-b).

Simpson's Index Values for Amphibians by Season

Spring	0.694
Summer	0
Fall	0.843

(a)

Simpson's Index Values for Reptiles by Season

Spring	0.906
Summer	0
Fall	0.725

(b)

Table 6a-b. Simpson's diversity index tables. The calculated Simpson's Diversity Indices for amphibians per season (a) and for reptiles per season(b) are arranged into tables.

Species evenness was also compared between trip or year and season. It was found that the community was most even in the survey in 2019, yet the evenness of all of the trips were relatively very close (Table 7a). The year with lowest species evenness was 2018 (Table 7a). The season with the most evenness between species was fall, showing significantly more evenness than the spring season did (Table 7b).

Species Evenness by Trip/Year

2018	0.875
2019	0.939
2021	0.899
2022	0.909
(a)	

Species Evenness by Season

Spring	0.437
Summer	0
Fall	0.708
(b)	

Table 7a-b. Species evenness indices. Species evenness as an index was calculated for trip or year (a) and for season (b) and arranged into tables.

Average air temperature in degrees C per trip indicated that the warmest trip was 2021 and the coolest 2018, yet all of these averages were close in number, especially between 2019 and 2022 (Table 8a). Average air temperatures per season indicated that the summer season was the warmest season while spring was the coolest on average (Table 8b).

Average Air Temperature (deg C) by Trip/Year

2018	18.7
2019	22.4
2021	25.5
2022	21.8
(a)	

Average Air Temperature (deg C) by Season

Spring	19.8
Summer	27.5
Fall	23.6
(b)	

Table 8a-b. Average temperatures. The average temperature for each trip or year (a) and season (b) were calculated by averaging the air temperatures at each capture over that timeframe.

Temperatures at detection for both types of herpetofauna were compared. The overall average temperature at detection for amphibians, considering both air and substrate temperatures, was determined to be 21.1 degrees C. The overall average temperature at detection for reptiles, considering both air and substrate, was determined to only be slightly higher, at 21.7 degrees C. The average air and substrate temperatures separately in degrees C for amphibian detections show that amphibians were detected at lower substrate temperatures than air temperatures on average (Table 9a). The average air and substrate temperatures separately in degrees C for reptilian detections show that reptiles were detected at very similar air and substrate

temperatures, but on average were detected at a slightly higher substrate temperature than air temperature (Table 9b). The amphibian average air temperature at detection was higher than this average for reptiles, however, the average substrate temperature at detection was lower for amphibians than it was for reptiles (Table 9a-b).

Average Amphibian Detection Temperature (deg C) by Temp. Type		Average Reptilian Detection Temperature (deg C) by Temp. Type	
Air	22.4	Air	21.5
Substrate	19.7	Substrate	21.9
(a)		(b)	

Table 9a-b. Average temperatures by herp. The average air and substrate temperature at detection for all amphibians captured in the study (a) and all reptiles captured in the study (b) were calculated by averaging the air or substrate temperatures at each capture over the entire survey.

The extremes of air and substrate temperature at detection for both amphibians and reptiles was also found and placed into tables for comparison. The coldest air temperature amphibian detection occurred at 14.7 degrees C, while the coldest substrate temperature amphibian detection occurred at 14.9 degrees C (Table 10a-b). The warmest air temperature amphibian detection occurred at 29.0 degrees C, while the warmest substrate temperature amphibian detection occurred at 22.8 degrees C (Table 10a-b).

Amphibian Detection Temperature Extremes

Air Temperature (deg C)		Substrate Temperature (deg C)	
Low	14.7	Low	14.9
High	29.0	High	22.8
(a)		(b)	

Table 10a-b. Extreme amphibian detection temperatures. The lowest and highest air temperature recorded at an amphibian detection (a) and lowest and highest substrate temperature recorded at an amphibian detection (b) are arranged in tables.

The coldest air temperature reptilian detection occurred at 14.2 degrees C, while the coldest substrate temperature reptilian detection occurred at 16.2 degrees C (Table 11a-b). The warmest

air temperature reptilian detection occurred at 28.1 degrees C while the warmest substrate temperature reptilian detection occurred at 31.4 degrees C (Table 11a-b).

Reptilian Detection Temperature Extremes

Air Temperature (deg C)		Substrate Temperature (deg C)	
Low	14.2	Low	16.2
High	28.1	High	31.4
(a)		(b)	

Table 11a-b. Extreme reptilian detection temperatures. The lowest and highest air temperature recorded at a reptilian detection (a) and lowest and highest substrate temperature recorded at a reptilian detection (b) are arranged in tables.

The coldest air temperature at detection was slightly lower for reptiles than for amphibians. The warmest air temperature at detection was also slightly lower for reptiles than for amphibians. The coldest substrate temperature at detection was higher for reptiles than for amphibians, as was the warmest substrate temperature at detection, but this second number was significantly higher than the amphibian maximum rather than just a degree or two.

Discussion

Again, the goal of this study was to detect and report the composition of herpetofauna species found in the study region of Northern Mississippi, and their diversity, abundance, dominance and rarity, richness, and evenness throughout the four separate survey trips in order to show trends in these measures over periods of time. The objective of this survey was to ultimately better inform the timing and methods for continuation and amplification of the survey in future years, to emphasize deviations from expectations for these parameters, give insight into the state of the herpetofauna community in the study region, create a baseline for comparison in continuation of the study, and note areas of conservational concern.

Expectations

Expectations for the outcomes of this survey were informed by broader herpetological distribution, ecology, life history characteristics, and presence in the particular study region. Firstly, while herpetofauna are distributed widely throughout the world, species prominence of both reptiles and amphibians is most notable in the warmer, more humid, temperate and tropical regions, particularly between 30 degrees N and 30 degrees S, although their ranges extend practically to every continent (Duellman and Zug 2022). The two groups only differ in that reptiles are usually more present in conditions such as extremely xeric deserts than amphibian species are (Duellman and Zug 2022). In the United States, this places the highest expected abundance and therefore variation of herpetological species in the desert, temperate, and subtropical ecosystems of the American south, southwest, and southeast (Olson and Saenz 2013, Valentine-Darby 2010). Mississippi in particular harbors a very diverse and abundant array of herpetofauna due to its location within that region of the country along with its diverse habitats throughout the state, affirming the site of this study as a prime location for such a herpetological survey into the future (Selman et al. 2018). Mississippi is known to have 30 endemic species of

amphibians and 84 endemic species of reptiles, putting it among the very top states by amount of herpetofauna and consistent with the countrywide amphibian and reptile ratio, with prior surveying confirming the high herpetofauna presence in this state (Selman et al. 2018).

The main drivers behind this global distribution of herpetofauna and expected abundance in the study region are some critical shared characteristics within the two groups of vertebrates. Herpetological species are ectothermic in nature, obtaining their heat from their environment rather than regulating body temperature internally. In addition, most species of herpetofauna are poikilothermic, using behavioral mechanisms to further regulate their temperature being obtained from their environment. All species of reptiles and amphibians have permeable skin as well. For these main ecological reasons, herpetofauna distribution overall is centered much more prominently around those warmer, wetter habitats and climates within the tropics, with distribution and variation lowering as you move away from those regions. These two groups require the aforementioned specific hygrothermal microclimates and environments within their habitats in order to develop, survive, and reproduce. These heat and moisture factors contribute to the expectation that more herpetofauna will be found in such regions as Northern Mississippi the most abundantly. These factors led to the expectation that the herpetological findings of this study would be robust, diverse, and will exhibit high presence of both amphibian and reptile species, with reptiles accounting for a larger proportion of detections, higher raw abundance, higher number of species, and also a wider diversity within those detections.

Further expectations are to be informed when addressing life history characteristics and ecology of amphibians and reptiles separately. Reptiles, unlike amphibians, have beta keratins within their epidermal cells, modifying them into scale structures on the outside of the skin, modifying their permeable skin to be significantly more resistant to desiccation than the much

more permeable and vulnerable skin of amphibian species. Some reptile species, namely turtles, possess a shell as well, also resisting the water loss to the environment which amphibian species are more at risk of. Further, reptiles produce amniotic eggs while amphibians do not. This life history characteristic allows the embryo of reptilian species to be protected and nourished within their own “aquatic environment” of water and energy sources contained within a shell, while still remaining permeable enough to allow the exchange of gases with the environment. The eggs of amniotes such as reptiles allows for the independence from proximity to and reliance on water sources for reproduction, leading to the increased ability of reptilian species to inhabit drier and warmer habitats. Amphibians, on the other hand are non-amniotes. Eggs of these species lack a shell and amniotic membranes, rather possessing several gelatinous layers surrounding the embryo, requiring an aquatic environment for embryos rather than having a self-contained one. Amphibian species have gills at birth, further relying upon water for gas exchange in the embryo and larval stages (Cossel 1997). Most species develop lungs later in their life cycles while some retain their larval gills and remain fully aquatic. As a result, amphibian species require a water source and proximity to water for their reproduction, survival, and continuation. Due to this difference in ecological and life history characteristics, amphibians are characterized as being more highly temporal species, and measures of detection and diversity for them are expected to vary more dependent on time of year, season, weather, and even time of day, due to changes to temperature and moisture, as those are two critical facets of their life history and behaviors which are less limiting on reptilian species.

This drastic difference between the two groups of herpetofauna is cause for different expectations concerning reptile and amphibian range, distribution, and detection in this study. The somewhat freeing of reptilian species from the dependence on water sources due to those

mentioned characteristics assumes that reptilian species have a wider habitat and range for condition tolerance than amphibians. This assumption forms the expectation that the two groups will vary in detection through the study largely based on temperature and time of year as indicators to moisture level available to them. Reptiles are expected to be detected and show more dominance, diversity, and abundance overall at hotter temperatures and in specific seasons which have warmer temperatures or drier conditions, while this would be expected for amphibians overall at lower temperatures and seasons with wetter conditions. Due to that difference in ecological and life history characteristics, amphibians are characterized as being more highly temporal species, and measures of detection and diversity for them are expected to vary more dependent on time of year, season, weather, and even time of day, due to changes to temperature and moisture, as those are two critical facets of their life history and behaviors which are less limiting on reptilian species.

While these overarching predictions exist based upon the major life history and ecological characteristics of these herpetofauna, expectations on detection and diversity arise also from other measures for this Northern Mississippi region in particular, such as species dominance and rarity. Predicted findings on overall herpetofauna and species abundance were informed by past studies and literature on Mississippi organisms. Snakes, lizards, skinks, and turtles are predicted to show the highest detection rates, abundance, and diversity in the study due to the dominance of these groups in the state of Mississippi overall and in the more northern regions specifically (Keiser 2018, Selman et al. 2018). There are almost triple the number of reptilian species in the state of Mississippi than the number of amphibian species, with the exact ratio of known reptilian to amphibian species being 2.8:1 (Keiser 2018). Within that group, snake species make up nearly half of all reptilian species, turtles make up about a third, and

lizards make up about a tenth, accounting for the higher predicted detection and diversity of these more common groups in the study region (Keiser 2018). Meanwhile, a significantly lower proportion of amphibian detections and diversity is expected, as the rarest species in the state are both amphibious: salamanders and hellbenders (Keiser 2018, Mississippi Museum of Natural Science 2014).

The meteorological averages of Northern Mississippi also informed predicted outcomes for the survey. May and June are the hottest months in the region with air temperatures maximizing around 31 degrees C and minimizing around 18.9 degrees C on average. October is close in temperature to mid-summer on average, with air temperatures maximizing around 24 degrees C and minimizing around 10 degrees C. This is similar to the month of April, having maximum temperature around 23 degrees C and minimum around 9 degrees C. These months are all warmer than the averages for March, where air temperatures maximize around 18 degrees C and minimize around 4 degrees C (NOAA 2023). March on average expects the most days of precipitation, 9. This is followed by May expecting 8, April and June expecting 7, and October expecting 5 days of precipitation on average (NOAA 2023). Given these historical records of climate and weather averages for the region along with knowledge on herpetological ecology and life history, reptile detection and diversity were expected to be highest in the trips occurring in the months of the summer season, and lowest in the spring season based on average monthly temperatures (NOAA 2023). On the other hand, amphibian detection and diversity are expected to maximize during the fall and spring season based on average monthly temperatures as well as a precipitation average in the spring which is double that of the summer and late summer months (NOAA 2023).

Outcomes

The outcomes of this herpetological survey were quite interesting and thought-provoking, providing information for timing and methods of future survey trips as well as calling for some explanation or assessment of potential bias in the study when compared to herpetological ecology, life history, and literature.

The findings of this study met those general distribution-based expectations. The high raw abundance, increased amount of distinct species or species richness, similar species evenness throughout time periods, and presence of herpetofauna species in this state due to its latitude and overall climate was observed. High herpetofauna presence and expected detection rates were met and confirmed by the sheer amount of distinct species found in just the relatively short survey period. The major expected classes of amphibians and reptiles were all detected, with species of salamanders and skinks found to be dominant as was supported in literature on the region (Olson and Saenz 2013, Selman et al. 2018, Valentine-Darby 2010). As found through abundance and proportion calculations of both species and amphibians and reptiles overall, the expectation that reptiles would outnumber detected amphibians in abundance as well as amount of species was supported. The found ratio of detected reptiles to amphibians, 1.39:1, was lower than expected for the state of Mississippi overall, yet still exhibited the prominence of reptilian over amphibian detection for the state and for the country.

Expectations on species presence, richness, dominance or rarity, and diversity overall and based upon season or climate were not all necessarily met. While herpetological species classes such as snakes, turtles, and lizards are the proven dominant species in the state and salamanders the rarest in the state, these classes did not end up exhibiting the expected dominance or rarity found throughout literature (Keiser 2018, Mississippi Museum of Natural Science 2014, Selman et al. 2018). Again, salamanders and skinks exhibited overall dominance, with the little brown

skink, the southern cricket frog, and the southern two-lined salamander found to be the dominant species from each trip year, and the Mississippi slimy salamander, the southern cricket frog, and the little brown skink dominated detections during both seasons – all three during spring, and the little brown skinks in the fall. This outcome lined up well with the expectations formed on presence of reptiles and amphibians in each season, as the amphibian salamander species showed dominance in the spring where it was expected due to lowered overall average temperatures, while the reptile species, the skink, was the most dominant by a significant margin in the fall with the higher expected and observed temperatures on average.

In comparing detection of reptiles and amphibians as classes, that overall ratio was as mentioned, a 1.39:1 favoring reptiles. However, the 2018 trip ended up showing the highest detection of amphibians, 2021 showed the highest detection of reptiles relative, and the other trips showed very close proportions relative. This observation did line up with expectations formed on seasonal climate averages as well as observed averages, as the amphibians were most commonly detected relative to the overall proportion in the cooler 2018 and 2022 season based on historical averages while reptiles were not most dominant in 2019 and 2021. These detections also lined up with the observed temperatures cooler in 2018 and 2022 and warmer in 2019 and 2021. As for season, the expectation that reptiles will be more prominent in the warmer and drier fall and the amphibians in the cooler, wetter spring, based on climate histories, was met, and this also lined up with the climate data collected during the survey as fall did prove to be warmer than spring. Summer was indeed the warmest, but the detection data from this season was an anomaly so it was left out of conclusions on season presence. Herpetofauna species richness data findings overall only partially lined up with expectations in this way as well. The highest amount of distinct species was detected in 2018 followed by 2022 and then the other two trips equally,

showing the most species detected in the trips with the lowest overall observed average temperatures. The spring also showed slightly more (two) distinct species than the significantly warmer fall, displaying a larger contrast in overall herpetofauna species detection that had been predicted based on seasonal temperatures. The species evenness findings partially lined up with expectations, showing the least amount of evenness in 2018 (caused by the higher detection of amphibian individuals here) and the highest amount of evenness in 2022, as across the board detections did not favor one group over the other for that trip. However, the evenness of the spring was significantly higher than the fall, stemming from the much higher presence of reptilian species such as the little brown skink in the fall, which dominated detections over a number of other reptilian and amphibian species in that season. This finding implied that higher temperatures may cause a skewed detection of certain species over others, making for a less even community detected.

Diversity was the next major measure, and expectations for this parameter were partially met by the findings concerning trip year and season. Overall, 2018 exhibited the highest amount of both Shannon-Weiner and Simpson's diversity out of the trips, while 2021 exhibited the least. Spring showed the slightly higher Shannon-Weiner and Simpson's diversity value than the fall did. This lines up with the higher abundance of species over all time periods, but only lines up with evenness findings in the seasons designation. The weighting of the abundances found in each particular trip outweighed the impact of evenness on these diversity calculations in that time period designation. This did not line up with the expectation that the community will be more diverse when the average climate is generally warmer due to herpetofaunal ectothermic nature. When the diversity, both Shannon-Weiner and Simpson's, of amphibians and reptiles was assessed separately, it was found that, overall, reptile species had much higher diversity values

across years than amphibians. It was found that while amphibians dominated abundance in 2018, reptiles had a significant amount, and actually the highest value found overall, of diversity in that trip. This met the expectation due to herpetofaunal distribution and the known herpetofaunal makeup of the state of Mississippi that reptilian species would be not only more detected, but also show wider diversity of species among those detections. On the other hand, amphibian diversity did end up outweighing reptilian in the fall season by a slight amount. This was not expected, as due to historical and observed climate data, it would be predicted that amphibian species detection in the community would narrow rather than expand during these warmer and some of the driest months of the year in this region due to the preferred cooler and wetter microclimates by this class.

When assessing temperature data related to reptilian and amphibian detections separately, it was found that the average air temperature at amphibian detection was slightly warmer than that for reptiles, while substrate temperature at detection was lower. This substrate average lined up well with expectations that amphibians will be detected at cooler temperatures, yet the air temperature average at detection did not. It was assumed that reptilian detection would yield a higher air and substrate temperature average, but this was not necessarily found to be true. However, when the summer anomaly of only salamanders detected was removed from the average temperature at detection findings for amphibians, this reduced the air temperature average at detection to almost the same as the reptilian average, making more sense and lining up better with expectations. When assessing temperature extremes at detection between the two types of herpetofauna, it was found that reptiles were actually detected at slightly lower air temperatures than amphibians. However, when that anomaly season was again removed, the maximum air temperature at an amphibian detection was lowered to 25.9 degrees C rather than

29 degrees C, again better lining up with expectations of cooler amphibian detections. The substrate temperature extremes at detection, however, lined up well with predictions, finding that reptiles were detected at a significantly higher maximum substrate temperature, and a higher minimum substrate temperature as well than amphibians.

Informing Survey Methods

The findings of this survey, again, are not only very intriguing when assessed compared to expectations on measures of these species and populations, but also are valuable information and a basis for planning on the timing and methods of future trips moving forward to also better contribute to this comprehensive survey goal.

It was found that a trip regardless of time of year will return more reptilian detections and distinct reptilian species than amphibian. A spring trip will result in likely relatively more amphibian detections than one in the late summer or fall would. When temperatures are cooler on average, the amphibian detection rate and abundance will also increase, while reptilian abundance will also increase, but stay more consistent over the seasons. A wider diversity of amphibian species will more likely be detected in the fall than the spring season, however, and a wider diversity of reptiles will be detected in the spring than fall season, both despite predictions based on known climate (moisture level) and temperature preferences for the groups. A spring trip will also return a likely slightly higher amount of distinct species across the board than a fall trip will, yet the evenness between species detected will maximize by a large margin in the fall and will be much more skewed towards certain species in the spring season. Trips in months with lower levels of rainfall will likely result in more reptilian detections, while trips in months with the higher amounts of rainfall showed a more similar proportion of amphibian and reptilian detection and abundance to the observed proportions for the survey over the entire study period, or the average proportions being detected by the survey.

The optimal temperature range for maximizing the detection of herpetofaunal presence and species diversity would fall between 18 degrees C and 32 degrees C, indicated by the study thus far. However, the specific optimal time of year for overall herpetofaunal detection is more difficult to distinguish and more multi-faceted. It is implied that a spring trip will give a better, more comprehensive view of both the amphibian and reptilian presence in the study region than a fall trip will, while a fall trip will give a better insight into actual diversity of amphibian species in the region than a spring trip will, yet the spring trip will maximize detection of more distinct reptilian species. As for informing timing of this trip, a fall trip is definitely valuable to the detection of more distinct species across the board, particular the more dominant species in the region such as skinks, salamanders, and snakes. A later spring trip, however, would also contribute massively to the goals of the survey, allowing the incorporation of some amphibian diversity and detection that would otherwise be absent from a purely fall trip and survey. A supplemental trip in the spring, maybe towards the beginning of summer, would greatly aid the goals of the study and would establish a much more accurate and comprehensive survey of the region. Since summer data is still largely unknown due to the skewed results coming from a directed study once in this season, a trip with student assistance in the summer would also be valuable in filling in a gap in the data which currently exists and would most likely maximize the detection and species diversity of reptiles in particular found in the study region. Overall, the most comprehensive and accurate survey will combine efforts in each season, due to the observed variability in species and broader herpetological types across times of the year which have not been found to consistently line up with expectations informed on known ecology, distribution, and life history characteristics of the organisms.

Possible Biases

Several potential biases have arisen as probable explanation for, or at least played a role in, the outcomes observed and for deviations from predicted detection and diversity based on known herpetological ecology and past literature findings.

Firstly, Mississippi has been in a drought state in the past decade, beginning in approximately 2010 (Rasmussen 2022, USGS 2022). The Mississippi River was found to be at its record historical low in 2022 (Rasmussen 2022, USGS 2022). This is clearly a notable source of potential bias in the current and ongoing species survey in the study region. Abundances, detection rates, and species diversity measurements across the board are all likely significantly lowered and reduced, relative to a normal, healthy ecosystem in this region, due to these impactful drought conditions and the sensitivity of herpetofauna to alterations and disturbances to their environments (Dodd 1992, Martensson 2016, Ruthven et al. 2002, USGS 2022, Whitfield Gibbons et al. 2000, Winter et al. 2013). Further, the amphibian rates are likely more significantly impacted by this ongoing climate condition than reptilian species are, due to those drastic differences in vulnerability to reduced moisture levels and to habitat disturbance of environmental water sources. Furthermore, while the seasons lined up with temperature predictions relative to one another, overall all seasons ended up averaging at either the upper end of their normal average range per month, or were hotter than that past average. This is likely another symptom of the ongoing drought conditions in the study region and has likely also similarly affected the presence and diversity of herpetofauna but amphibian species more in particular, narrowing a likely more abundance and diverse community to a restricted and more threatened number. With both of these biases comes the difficulty to fully affirm this assumption, as the herpetological survey did not extend to before this drought and change to environmental

temperatures began in the region, so it is difficult to compare findings with what should be observed in a healthier, balanced ecosystem.

This outstanding source of potential bias in the study feeds directly into another larger source of bias in herpetological surveys worldwide in recent years: the issue of climate change. As herpetological species are, again, highly sensitive to altered temperatures and moisture levels, particularly in more temperate and subtropical regions instead of more fully tropical zones, the changes to Mississippi climate increasing with each passing year are definitely impacting the species and their abundances detected, as well as the overall composition of the Northern Mississippi study region (Olson and Saenz 2013, Rasmussen 2022, USGS 2022). Reproduction for herpetofauna are highly related to close time windows in specific seasons and condition ranges, with warmer spells for too long of a period of time having proven fatal across the board species-wise (Olson and Saenz 2013, Rasmussen 2022, USGS 2022). Altered thermal niches due to the changes to climate of the region are likely making the reproductive windows or range of suitable conditions of Mississippi and global herpetofauna successful breeding significantly narrower, skewing the results of such surveys on their populations and communities (Olson and Saenz 2013, Rasmussen 2022, USGS 2022). The increased agricultural intensification rates observed throughout the state of Mississippi are surely another source of habitat and conditions disturbance, posing a similar detriment to herpetological diversity in the study region and lower United States overall (Leyte-Manrique et al. 2019, USGS 2022).

Further, methods of this survey in particular over the years thus far pose another source of bias to its findings. The visual encounters and by-hand disturbances and trappings are largely based upon human discretion in the trips thus far. Not all seasons of the year, summer in particular, are evenly represented in this survey, despite the known variations in detection and

diversity that would be observed throughout a comprehensive, full-year study. An entire year was left out of this study due to the pandemic, 2020, and this gap in the data is likely a cause for making overall trends much more difficult to detect and to then compare successfully to future surveying in the region. The weather or precipitation amount was not taken into account or recorded for captures during this trip; it's known that there was much more rainfall during specific trips than others, but this factor was not incorporated into the findings or discussion of the findings of this survey thus far, potentially skewing conclusions to be made on the impact of not only climate but more local weather conditions on the detection and diversity of herpetofaunal species in the study region. In addition, exact proximity to water bodies was not measured either or taken into account in any form in this survey thus far, possibly skewing detection toward reptilian and away from amphibian species due to an uneven amount of habitats or separate microclimates surveyed.

Survey Improvements

Clearly, numerous improvements can be made to this survey in trips going forward to better the detection and representation of all species actually present in the region. Additional in-situ methods such as measuring and recording proximity to water for trap sites or hand-trapping sites, noting slope direction of detections in appropriate landscapes, and noting as well canopy cover, shade, or available sunlight factor would all lead to more comprehensive and accurate findings on the conditions for amphibians and reptiles as broader groups but also for differences in detection and diversity at the species level as well. Counting all individuals detected instead of just ones which had thorough examinations and measurements performed on them will further significantly improve the comprehensiveness of the study. Practicing consistency with laying or placing traps or detection sites throughout the study area by implementing a more thorough and organized pattern for trap sites, such as a grid fashion across the region, as well as increased

amounts of trap sites overall, will greatly improve the success of detections along with reducing potential biases in trap site placements. In addition, classifying weather conditions and potentially substrate moisture level for each detection as opposed to only temperatures will further improve the survey and the quality of its findings and conclusions on herpetological presence.

Additional ex-situ methods of data analysis, such as performing Chi-squared tests on specific species, will again return better conclusions on the presence of species relative to one another and to the entire study region. Adjusting diversity measurements for species importance is another method and measurement more recently incorporated into such species surveys (Ontoy and Padua 2014). By including a weighted value for importance of species to their community, informed, derived, and calculated by domains deemed important to conservation biology such as species endemism, economic utility, functional role in the ecosystem, and IUCN risk status of the species (threatened or endangered), diversity indices such as Shannon-Weiner and Simpson's are enhanced, with simulations in past studies confirming that a new index such as this successfully aids in conservation decisions particularly in cases where the Shannon's indices of the ecosystems are equal or near equal, or in situations where the Shannon's index clearly identifies a site but the relative importance of the species found in other sites is heavier (Ontoy and Padua 2014). This would be a very worthwhile inclusion and expansion for our ongoing survey in Northern Mississippi, assisting further in the accurate representation of not only the species but to areas of particular concern and to giving a more solid baseline for conservation awareness, attention, efforts, and prioritization going into the future.

Importance to Conservation

This study has value moving into the future, especially concerning the changing status of herpetofauna globally. Surveys incorporating diversity indices and other measures of species composition in target regions provide important information about composition and structure of communities which holds particular value for the field of conservation biology. The ability to quantify diversity and species composition in this way has proven a critical tool for biologists in understanding community structure related to larger biodiversity conservation goals.

Such studies as this survey, employing species diversity and related measures, have become major components to conservation planning more recently in the realm of ecology (Dodd 1992, Iknayan et al. 2014, Laurencio and Fitzgerald 2010, Moritz et al. 2005, Winter et al. 2013). Monitoring of species diversity and the trends of other species measures are now fundamental for wildlife conservation, becoming abundantly more relevant to the field of conservation biology and have proven critical for evaluating the effectiveness of past management efforts as well as informing future management decisions (Baumgart, Martensson 2016, Moritz et al. 2005, Winter et al. 2013). Species diversity indices estimate variation of species within and among systems, prioritizing species and habitat, informing actions and effort, and bridging research and actual conservation (Martensson 2016, Moritz et al. 2005, Winter et al. 2013). Conservation planning in multiple instances globally and across the board species-wise has been accelerated and made more effective due to the incorporation of species diversity assessment and trends in related species measures (Martensson 2016, Moritz et al. 2005, Rabbe et al. 2022). Diversity indices – these mathematical models calculating number of species, distribution evenness, and species dominance with weighting in a particular community or compared across multiple have proven to inform management decisions in reducing overall biodiversity loss, overcoming difficult conservation targets, evaluating spatial diversity patterns,

outlining detection, assessing levels and magnitude of disturbance, and prioritizing efforts to the appropriate species or habitats (Baumgardt et al. 2021, Gering et al. 2003, Ghosh and Basu 2020, Iknayan et al. 2014, Leyte-Manrique et al. 2019, Moritz et al. 2005). Herpetofauna have been one of the most prominent targets of such species diversity-informed conservation tactics globally as of late, with particular use and emphasis on species diversity indices to accurately assess composition, trends over time, conservation threat status, extent of habitat disturbance, and to inform species conservation efforts moving forward (Baumgardt et al. 2021, Dodd 1992, Ghosh and Basu 2020, Laurencio and Fitzgerald 2010, Leyte-Manrique et al. 2019, Moritz et al. 2005, Rabbe et al. 2022, Russell et al. 2002, Ruthven et al. 2002). The value of this extended study thus far on a quite compromised group of organisms will prove critical in moving forward in their conservation, particularly with attention to bias and improvements to methods found in the four years incorporated moving into the future.

Literature Cited

- Baumgardt J, Morrison M, Brennan L, Thornley M, Campbell T. 2021. Variation in herpetofauna detection probabilities: implications for study design. *PMC. Environ Monit Assess.* 193(10): 658.
- Beals M, Gross L, Harrell S. 1999. Diversity indices: Simpson's D and E. NIMBIOS. [accessed 9 May 2023].
<http://www.nimbios.org/~gross/bioed/bealsmodules/simpsonDI.html#:~:text=Introduction%3A%20A%20diversity%20index%20is,of%20species%20in%20a%20community.>
- Behangana M, Magala R, Katumba R, Ochanda D, Kigoolo S, Mutebi S, Dendi D, Luiselli L, Hughes D. 2020. Herpetofaunal diversity and community structure in the Murchinson Falls-Albert Delta Ramsar site, Uganda. *European Journal of Ecology.* 6(2): 1-17.
- Briones Jr. S. 2017. Re: What is the deference between Shannon-Wiener diversity index and Simpson diversity Index? ResearchGate. [accessed 9 May 2023].
<https://www.researchgate.net/post/What-is-the-deference-between-Shannon-Wiener-diversity-Index-and-Simpson-diversity-Index/59159ea9f7b67ec86c7e4432/citation/download.>
- Cossel J. 1997. The class amphibia. DigitalAtlas. ISU. [accessed 9 May 2023].
<https://digitalatlas.cose.isu.edu/bio/amph/main/clasamph.htm#:~:text=Amphibians%20lay%20eggs%20that%20are,also%20important%20in%20gas%20exchange.>
- Dodd K. 1992. Biological diversity of a temporary pond herpetofauna in north Florida sandhills. National Ecology Research Center. *Biodiversity and Conservation.* 1: 125-142.
- Ekberzade B. 2022. Re: Does Shannon index represent largest diversity index when the individual belongs to different species and vice versa? ResearchGate. [accessed 9 May 2023].
https://www.researchgate.net/post/Does_Shannon_index_represent_largest_diversity_index_when_the_individual_belongs_to_different_species_and_vice_versa/62ea7f00694b9ef3f70d3daf/citation/download.
- Gering J, Crist T, Veech J. 2003. Additive partitioning of species diversity across multiple spatial scales: implications for regional conservation of biodiversity. *Society for Conservation Biology.* 17(2): 488-499.
- Ghosh D, Basu P. 2020. Factors influencing herpetofauna abundance and diversity in a tropical agricultural landscape mosaic. *Assoc. for Tropical Biology and Conservation.* BioTropica.
- Gibbons JW, Stangel PW. 1999. Conserving amphibians and reptiles in the new millenium. Partners in Amphibian and Reptile Conservation (PARC). Savannah River Ecology Laboratory.
- Grant B, Tucker A, Lovich J, Mills A. 1992. The use of coverboards in estimating patterns of reptile and amphibian biodiversity. *Wildlife 2001: Populations.* ResearchGate. 379-403.

- Iknayan K, Tingley M, Furnas B, Bessiniger S. 2014. Detecting diversity: emerging methods to estimate species diversity. ScienceDirect. Trends in Ecology and Evolution. 29(2): 97-106.
- Keiser E. 2018. Amphibians. Center for Study of Southern Culture. The Mississippi Encyclopedia. [accessed 9 May 2023].
<https://mississippiencyclopedia.org/entries/amphibians/#:~:text=Mississippi%20has%20thirty%20species%20representing%20seven%20families>.
- Kumar P, Dobriyal M, Kale A, Pandey A, Tomar R, Thounaojam E. 2022. Calculating forest species diversity with information-theory based indices using sentinel-2A sensor's of Mahavir Swami wildlife sanctuary. NIH. PloS One. 17(5).
- Kusamka C. 2019. Using diversity indices for identifying the priority sites for herpetofauna conservation in the Democratic Republic of the Congo. CyberLeninka. Research Center in Natural Sciences at Lwiro, Democratic Republic of the Congo. 4(3): 13-33.
- Laurencio D, Fitzgerald L. 2010. Environmental correlates of herpetofaunal diversity in Costa Rica. Journal of Tropical Ecology. 26(5): 521-531.
- Leyte-Manrique A, Abel Antonio BC, Maciel-Mata CA. 2019. A comparison of amphibian and reptile diversity between disturbed and undisturbed environments of Salvatierra, Guanajuato, Mexico. Sage Journals Tropical Conservation Science. 12.
- Martensson R. 2016. Species and biological diversity: choices of diversity indices and their potential consequences for nature conservation. Lunds Universitet. 2(15).
- Mississippi Museum of Natural Science. 2014. Endangered species of Mississippi. Mississippi Department of Wildlife, Fisheries, and Parks. Mississippi Museum of Natural Science. [accessed 9 May 2023]. <https://www.mdwfp.com/media/301899/this-is-the-print-version-of-endangered-species-of-mississippi-as-it-went-to-print-may-through-june-2014.pdf>.
- Mohammed M. 2021. Re: Which biodiversity index is more reliable/better and how can it be applied to a mixed vegetation? How might I differentiate diversity indices and IVI? ResearchGate. [accessed 9 May 2023]. <https://www.researchgate.net/post/Which-biodiversity-index-is-more-reliable-better-and-how-can-it-be-applied-to-a-mixed-vegetation-How-might-I-differentiate-diversity-indices-and-IVI/61041345f3afa9132666756d/citation/download>.
- Moritz C, Hoskin C, Graham C, Hugall A, Mousalli A. 2005. Historical biogeography, diversity, and conservation of Australia's tropical rainforest herpetofauna. Phylogeny and Conservation. Cambridge University Press. 10: 243-264.
- NOAA. 2023. Mississippi weather report archives. NCEI. [accessed 9 May 2023].
<https://www.ncei.noaa.gov/>.
- Nolan K, Callahan J. 2006. Beachcomber biology: the Shannon-Weiner species diversity index. Conference of the Association for Biology Laboratory. Tested Studies for Laboratory Teaching. 27: 334-338.
- Olson D, Saenz D. 2013. Climate change and reptiles. U.S. Department of Agriculture, Forest Service, Climate Change Resource Center. [accessed 9 May 2023].

- <https://www.fs.usda.gov/ccrc/topics/reptiles-and-climate-change#:~:text=The%20highest%20biodiversity%20of%20reptiles,considerably%20as%20you%20go%20north.>
- Ontoy D, Padua R. 2014. Measuring species diversity for conservation biology: incorporating social and ecological importance of species. Cebu Normal University. Biodiversity Journal. 5(3): 387-390.
- Rabbe F, Jaman F, Alam M, Rahman M, Sarker M. 2022. Species diversity, composition, and distribution of the herpetofauna in the northwestern region of Baangladesh. University of Dhaka Department of Zoology. Amphibian and Reptile Conservation. 16(1): 226-234.
- Rain R. 2022. Shannon diversity index. [accessed 9 May 2023]. <https://www.omnicalculator.com/ecology/shannon-index>.
- Rasmussen C. 2022. Why the Mississippi river basin just had its worst drought in 1,200 years. AZ Animals. [accessed 10 May 2023]. <https://a-z-animals.com/blog/why-the-mississippi-river-basin-just-had-its-worst-drought-in-1200-years/>.
- Ruthven D, Kazmaier R, Gallagher J, Synatzske. 2002. Seasonal variation in herpetofauna abundance and diversity in the south Texas plains. Oklahoma State University, Department of Biology. The Southwestern Naturalist. 47(1): 102-108.
- Selman W, Aughtman A, Anderson A, Carter D. 2018. Thirty-six new county records for herpetofauna of Mississippi, Louisiana, and Tennessee, USA: results from an undergraduate herpetology class assignment. Herpetological Review. 49(1): 85-88.
- Smith E, Berry E. 2010. Comparing diversity indices using field data from a riparian New Hampshire forest community. Saint Anselm College Department of Biology.
- USGS. 2022. Why are amphibian populations declining? USGS. [accessed 9 May 2023]. <https://www.usgs.gov/faqs/why-are-amphibian-populations-declining>.
- Urbina-Cardona J. 2008. Conservation of neotropical herpetofauna: research trends and challenges. Facultad de Ciencias, UNAM. Mongabay Open Access Journal Tropical Conservation Science. 1(4): 359-375.
- Valentine-Darby P. 2010. Reptiles and amphibians – introduction, distribution, and life history. Southern Plains Network Inventory and Monitoring Program. National Park Service. [accessed 9 May 2023]. <https://www.nps.gov/articles/reptiles-and-amphibians-distribution.htm#:~:text=There%20are%20about%20460%20species,at%20least%2020%20amphibian%20species.>
- Whitfield Gibbons J, Scott D, Ryan T, Buhlmann K, Tuberville T, Metts B, Greene J, Mills T, Leiden Y, Poppy S, Winne C. 2000. The global decline of reptiles, déjà vu amphibians: reptile species are declining on a global scale, six significant threats to reptile populations are habitat loss and degradation, introduced invasive species, environmental pollution, disease, unsustainable use, and global climate change. BioScience. 50(8): 653-666.
- Wilson A, Gownaris N. 2022. Diversity indices. LibreTexts Biology. [accessed 9 May 2023]. https://bio.libretexts.org/Courses/Gettysburg_College/01%3A_Ecology_for_All/22%3A_Biodiversity/22.02%3A.

Winter M, Devictor V, Schweiger O. 2013. Phylogenetic diversity and nature conservation: where are we? ScienceDirect. Trends in Ecology and Evolution. 28(4): 199-204.